

TEST_STRATEGY

HelixTrack Test Strategy — 2026-06-21

Overview

Comprehensive test strategy for HelixTrack across all platforms and components.

Test Types

1. Unit Tests

- **Purpose:** Test individual components/functions in isolation
- **Tools:**
 - Backend: Go testing package
 - Web: Jasmine + Karma
 - Desktop: Jasmine + Karma
 - Android: JUnit + Mockito
 - iOS: XCTest
- **Coverage Target:** 80%+ for all platforms

2. Integration Tests

- **Purpose:** Test component interactions
- **Tools:**
 - Backend: Go testing + test containers
 - Web: Angular TestBed
 - Desktop: Angular TestBed
 - Android: Espresso
 - iOS: XCUITest
- **Coverage Target:** 60%+ for critical paths

3. E2E Tests

- **Purpose:** Test complete user workflows
- **Tools:**
 - Web: Cypress + Playwright
 - Desktop: Tauri E2E

- Android: Espresso
- iOS: XCUITest
- **Coverage Target:** Critical user journeys

4. Performance Tests

- **Purpose:** Test system performance under load
- **Tools:**
 - Backend: Go benchmark tests
 - Web: Lighthouse + Web Vitals
 - Desktop: Tauri performance profiling
 - Android: Android Profiler
 - iOS: Instruments
- **Targets:**
 - API response time: < 200ms
 - Page load: < 2s
 - Frame rate: 60fps

5. Security Tests

- **Purpose:** Test security vulnerabilities
- **Tools:**
 - SAST: Go security scanners
 - DAST: OWASP ZAP
 - Dependency: npm audit, go mod verify
- **Coverage:** All public endpoints

6. Accessibility Tests

- **Purpose:** Test WCAG compliance
- **Tools:**
 - Web: axe-playwright
 - Desktop: axe-playwright
 - Android: Accessibility Scanner
 - iOS: Accessibility Inspector
- **Target:** WCAG 2.1 AA compliance

Test Matrix

Component	Unit	Integration	E2E	Performance	Security	Accessibility
Core Backend	✓	✓	✓	✓	✓	N/A
Web Client	✓	✓	✓	✓	✓	✓
Desktop Client	✓	✓	✓	✓	✓	✓
Android Client	✓	✓	✓	✓	✓	✓

Component	Unit	Integration	E2E	Performance	Security	Accessibility
iOS Client	✓	✓	✓	✓	✓	✓

Test Automation

CI/CD Integration

- **Backend:** Go test + coverage
- **Web:** ng test + ng e2e
- **Desktop:** ng test + Tauri E2E
- **Android:** gradle test + gradle connectedAndroidTest
- **iOS:** xcodebuild test

Test Scripts

```
# Backend
cd core/Application && go test ./...

# Web Client
cd web_client && npm test && npm run test:e2e

# Desktop Client
cd desktop_client && npm test && npm run test:e2e

# Android Client
cd android_client && ./gradlew test && ./gradlew
    connectedAndroidTest

# iOS Client
cd ios_client && xcodebuild test
```

Test Data Management

Test Fixtures

- Backend: SQL seed files
- Web: JSON fixtures
- Desktop: JSON fixtures
- Android: JSON fixtures
- iOS: JSON fixtures

Test Databases

- Backend: SQLite in-memory
- Web: Mock services
- Desktop: Mock services

- Android: Room in-memory
 - iOS: GRDB in-memory
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Test Reporting

Coverage Reports

- Backend: go tool cover
- Web: Istanbul
- Desktop: Istanbul
- Android: JaCoCo
- iOS: Xcode coverage

Test Results

- Backend: go test -v
 - Web: Karma reporters
 - Desktop: Karma reporters
 - Android: Android test reports
 - iOS: Xcode test reports
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Test Execution

Local Development

```
# Run all tests  
make test-all
```

```
# Run specific platform  
make test-backend  
make test-web  
make test-desktop  
make test-android  
make test-ios
```

CI/CD Pipeline

- Pre-commit: Lint + format check
 - Pre-push: Unit tests
 - Pull request: Full test suite
 - Merge: Integration tests
 - Release: E2E + performance + security
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Test Maintenance

Regular Tasks

- Update test fixtures
- Review test coverage
- Fix flaky tests
- Update test documentation

Test Debt Tracking

- Track failing tests
 - Monitor coverage trends
 - Review test quality
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Cross-references

- Implementation Plan
- Gap Analysis
- Architecture

Sources verified 2026-06-22: /Volumes/T7/Projects/helix_track/ (sibling HelixTrack source tree; core test suites + web_client cypress) , <https://github.com/Helix-Track/Everything>

REPO-STATE-DERIVED (per §11.4.99 the sources are the cross-referenced HelixTrack source tree, following the docs/ARCHITECTURE.md precedent). Cross-referenced on 2026-06-22: - The test strategy targets the present sibling HelixTrack tree; real test scaffolding is present (e.g. helix_track/desktop_client/cypress/ + cypress.config.ts, core/*test* suites) confirming the cross-platform test surface this strategy plans for. - This is a project-internal STRATEGY for HelixTrack, not external-tool instructions; it asserts no third-party-tool version that needs staleness-checking. Re-baseline coverage targets against the live test suites as they grow.